

TITLE

Introduction

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Problem Statement

Retrieval-Augmented Generation (RAG) has become a foundational mechanism for grounding Large Language Models (LLMs) in external knowledge. However, existing frameworks continue to face two persistent challenges: semantic misalignment caused by domain-specific terminology and performance instability resulting from excessive contextual integration (An et al., 2024). Although recent agentic approaches introduce advanced augmentation and global context strategies, these methods often prioritize absolute accuracy without systematically evaluating their computational cost. As a result, the relationship between model scale, terminological methodology, and resource efficiency remains insufficiently understood.

Research Statement and Objectives

This study proposes the Assistant Researcher AI (ARAI), a tunable agentic framework designed to optimize the balance between structural coherence and resource efficiency in retrieval-augmented environments. By implementing a dual-flow architecture—comprising an Offline-Indexing Flow (OIF) for sectional stabilization and an Online-Retrieval Flow (ORF) for terminological refinement—this work investigates how

different terminological methodologies interact with model scale under constrained resource budgets.

The primary objective is to evaluate the impact of evidence-based, explanation-based, and hybrid methodologies on generation performance (ROUGE-L) and resource efficiency across three LLM architectures.

Research Questions

Primary Research Question

To what extent does aligning terminological methodology with model scale influence the balance between generation quality and resource efficiency in retrieval-augmented systems?

Supporting Research Questions

- Comparative: How does model scale affect the optimal form of terminological conditioning required to maximize generation performance?
- Efficiency-Oriented: Can targeted terminological intervention achieve competitive performance while reducing contextual overhead compared to standard retrieval-augmented approaches?
- Capacity-Oriented: At what point does increasing contextual integration cease to produce meaningful performance gains?

Literature Review

Do it later, find papers from 2025 onward make it at least 10; add background knowledge i.e. BART, LLM

Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) was introduced by Lewis et al. (2021) to mitigate the limitations of parametric-only knowledge in large language models (LLMs), where factual information is implicitly encoded in model weights and updating knowledge typically requires additional training. Instead of modifying parameters to incorporate new knowledge, RAG augments generation with non-parametric evidence retrieved from an external corpus. The framework couples a dual-encoder dense retriever, which maps queries and documents into a shared embedding space for similarity-based retrieval, with an encoder–decoder Transformer generator (BART). In the sequence formulation, the top-k retrieved documents are concatenated with the input query to condition the decoding process. By combining parametric representations with dynamically retrieved external evidence, RAG improves factual grounding and performance on knowledge-intensive tasks compared to standalone LLMs. Because generation is explicitly conditioned on retrieved documents, retrieval quality serves as a critical determinant of downstream performance.

RAG with Long-Context LLMs

Because retrieval quality critically influences downstream generation, recent advances in long-context LLMs have prompted renewed investigation into how expanded context windows interact with retrieval-augmented architectures. Rather than replacing retrieval, long-context models enable RAG systems to incorporate larger volumes of retrieved evidence

within a single forward pass, potentially reshaping the trade-offs between retrieval precision, context capacity, and computational efficiency.

Empirical analysis by Jin et al. (2024) reveals that increasing the number of retrieved passages in long-context RAG does not produce monotonic performance gains. While expanding the retrieval set initially improves answer quality through higher recall, performance subsequently deteriorates beyond a threshold. The authors attribute this degradation primarily to the accumulation of “hard negatives” — semantically similar but irrelevant passages that introduce noise into extended contexts. To mitigate this effect, they propose both training-free retrieval reordering strategies that exploit positional biases and training-based approaches, including implicit fine-tuning on noisy inputs and supervised intermediate-reasoning fine-tuning to improve relevance discrimination. These findings indicate that long-context capacity alone does not resolve retrieval-induced noise.

Complementing this mechanistic analysis, Leng et al. (2024) conduct a large-scale evaluation across 20 commercial and open-source LLMs, systematically varying total context length from 2,000 to 128,000 tokens (and beyond when supported). Their results show that although retrieving more documents can initially improve performance, only a small subset of state-of-the-art models maintain stable accuracy at extended contexts above 64k tokens. Performance frequently plateaus and declines beyond model-specific limits, with distinct failure modes emerging under long-context conditions.

Collectively, these studies demonstrate that RAG performance under expanded context windows exhibits diminishing returns and model-dependent task saturation thresholds. Increasing context capacity does not guarantee improved outcomes; instead, performance is governed by the balance between evidence coverage and contextual noise, highlighting the need for retrieval strategies that optimize not only recall but also information density and efficiency.

Parametric RAG and Knowledge Integration Beyond Context Expansion

While prior work has focused on optimizing retrieval quality and managing long-context limitations within input-augmented architectures, Su et al. advance a complementary line of inquiry by questioning the fundamental efficacy of in-context retrieval itself. They argue that expanding the input sequence with multiple retrieved documents not only increases computational overhead and inference latency, but can also degrade performance in tasks requiring complex reasoning. Beyond efficiency concerns, the authors identify a structural limitation of conventional RAG: retrieval-augmented methods inject knowledge at the input level, whereas large language models encode and operationalize knowledge primarily within their parameters. Consequently, input-level augmentation may constrain the depth of knowledge integration achievable during inference.

To address these limitations, Su et al. propose Parametric RAG, a paradigm that embeds external knowledge directly into the model’s parameter space rather than its context window. Instead of appending retrieved documents as textual input, the framework converts each document into a lightweight, document-specific parameter update applied to the feed-forward layers of the LLM. This document parameterization is achieved through a two-stage process. First, during an offline indexing phase, documents undergo augmentation via iterative rewriting and question–answer pair generation to promote robust knowledge abstraction. Second, a parameter-efficient adaptation technique (RoLA) encodes the augmented document into low-rank parameter matrices associated with the model’s feed-forward networks.

At inference time, the system follows a Retrieve–Update–Generate pipeline. After retrieving the top-k relevant documents, it loads their corresponding low-rank parameter

updates and merges them with the frozen base model weights to form a document-conditioned model. Response generation then proceeds without injecting lengthy textual evidence into the context window. By shifting knowledge integration from the input channel to the parametric channel, the framework reduces online computational cost while enabling deeper incorporation of external information.

Empirical results demonstrate that Parametric RAG consistently outperforms conventional in-context RAG across multiple LLM architectures in both effectiveness and efficiency. Moreover, the authors show that parametric integration is complementary to traditional retrieval augmentation, with hybrid configurations yielding further gains. Collectively, these findings suggest that scaling context length alone does not resolve retrieval-induced limitations; instead, architectural strategies that internalize external knowledge within model parameters represent a promising direction for advancing retrieval-augmented systems.

Semantic Mismatch and Domain-Specific Terminology

A persistent challenge in retrieval-augmented systems arises from semantic mismatch between user queries and domain-specific corpora, particularly when queries contain jargon, abbreviations, or context-dependent terminology. Standard dense retrievers often struggle in these environments because embedding similarity alone may fail to resolve ambiguous or underspecified domain expressions, leading to suboptimal retrieval.

To address this issue, An et al. (2024) proposed Golden-Retriever, an agentic RAG framework that introduces a reflection-based query augmentation stage prior to retrieval. Instead of directly embedding the raw user query, the system first identifies domain-specific jargon and abbreviations, determines their contextual meaning using a predefined taxonomy, and retrieves extended definitions from a domain dictionary. The query is then augmented

with these clarifications before being passed to the retriever. By enriching the semantic representation of the query, Golden-Retriever reduces ambiguity and improves alignment with industrial knowledge bases.

Empirical evaluation on a domain-specific question-answer dataset using three open-source LLMs demonstrates significant gains in retrieval and answer accuracy compared to traditional RAG pipelines. These findings suggest that retrieval failures in specialized domains often stem not from insufficient context length, but from semantic under-specification at the query level.

Existing research on retrieval-augmented generation has identified two primary performance constraints: degradation under extended context due to retrieval noise, and semantic mismatch caused by domain-specific terminology. While these studies clarify important failure modes, they largely evaluate performance under fixed methodological assumptions and prioritize accuracy as the dominant metric. Less attention has been given to how different terminological conditioning strategies interact with model scale and resource efficiency constraints.

Summary of Research Gaps

Lack of Methodological Differentiation in Terminological Conditioning

Prior RAG studies typically vary retrieval size, context length, or model architecture, but treat retrieved content as homogeneous evidence. Limited work systematically compares alternative terminological methodologies—such as evidence-based retrieval, explanation-based augmentation, or hybrid approaches—to determine how different forms of contextual support influence downstream generation. As a result, it remains unclear which conditioning strategy best balances coherence and efficiency across models of varying scale.

Insufficient Examination of Efficiency–Performance Trade-offs

Although recent studies report diminishing returns and task saturation under extended contexts, performance is primarily measured using accuracy-based metrics. The interaction between generation quality and token consumption remains underexplored. In particular, few works quantify how resource reduction (e.g., token budget minimization) affects performance stability. This leaves unresolved the question of whether optimal RAG configurations should be evaluated solely on output quality or on efficiency-adjusted performance.

Limited Analysis of Model-Scale Sensitivity

Existing evaluations compare multiple LLMs but rarely analyze how optimal retrieval or conditioning strategies vary as a function of model scale. Larger models may tolerate contextual noise differently from mid-scale architectures, suggesting that retrieval and terminological interventions may not transfer uniformly across models. A systematic investigation into parameter–method alignment across architectures remains absent.

Addressing these gaps requires a framework that explicitly models the relationship between terminological conditioning, model scale, and resource allocation. This study proposes the Assistant Researcher AI (ARAI), a tunable agentic architecture designed to evaluate parameter–method optimization under efficiency constraints.

Problem Formulation and Research Objectives

Problem Formulation

Retrieval-augmented generation (RAG) systems must balance generation quality with computational resource consumption. While prior studies have examined retrieval size, context length, and semantic alignment independently, limited work has formalized how different terminological conditioning strategies interact with model scale under constrained token budgets.

We frame this as a parameter–method optimization problem:

Given a model of scale S and a terminological conditioning strategy M , determine the configuration that maximizes generation performance while minimizing resource utilization.

In this formulation:

- S represents model architecture and parameter scale (e.g., 11B vs 17B).
- M represents terminological methodology (evidence-based retrieval, explanation-based augmentation, or hybrid configuration).
- Performance is evaluated using ROUGE-L.
- Resource utilization is quantified through a tokenizer-agnostic word-based proxy.

The objective is not to maximize performance in isolation, but to identify the configuration at which performance gains remain proportional to token expenditure before entering a saturation regime.

Conceptual Definitions

To operationalize this investigation, we define the following constructs.

Token Reduction Rate (TRR)

Token Reduction Rate (TRR) measures the relative change in resource consumption compared to a standardized retrieval-augmented baseline. It reflects efficiency improvement rather than raw performance gain. The formal computation is detailed in Methodology section.

Task Saturation

Task Saturation refers to the regime in which increasing contextual input or methodological complexity fails to produce proportional improvements in evaluation metrics, often resulting in performance plateau or degradation. It captures the diminishing marginal utility of additional context or evidence.

Efficiency Apex

Efficiency Apex denotes the configuration at which the ratio between generation performance and resource consumption is maximized prior to the onset of Task Saturation.

Conceptually, it represents the optimal balance point between quality and efficiency for a given model scale.

Research Objectives

This study aims to:

- Evaluate how evidence-based, explanation-based, and hybrid terminological methodologies influence ROUGE-L performance across varying model scales.
- Quantify the trade-off between performance and resource efficiency using TRR.
- Identify model-specific Efficiency Apex points to determine optimal parameter–method alignment in retrieval-augmented environments.

Research Questions

Primary Research Question

To what extent does strategic alignment between terminological methodology and model scale determine the location of the Efficiency Apex before Task Saturation occurs?

Supporting Research Questions

RQ1 (Comparative).

How does model scale influence the optimal terminological conditioning strategy required to maximize ROUGE-L without entering a saturation regime?

RQ2 (Evaluation-Oriented).

Can standalone terminological intervention (FF configuration) achieve superior resource return-on-investment compared to retrieval-augmented configurations (TF/TT) when evaluated through both ROUGE-L and TRR?

RQ3 (Mechanistic)

At what level of contextual integration does marginal performance gain approach zero or become negative, indicating the onset of Task Saturation?

Methodology

The Assistant Researcher AI (ARAI) system employs two distinct agentic workflows: the Offline-Indexing Flow (OIF) and the Online-Retrieval Flow (ORF). The OIF operates upon document ingestion into the system, functioning as a comprehensive document knowledge management framework that encompasses document registration, data preprocessing, and knowledge storage processes. Conversely, the ORF is activated when users seek to acquire knowledge or comprehension regarding specific documents. The ORF constitutes a knowledge retrieval mechanism that encompasses user query processing, terminology extraction, contextual retrieval, and response generation.

Offline-Indexing Flow (OIF)

The OIF architecture incorporates three primary storage components: DocumentMemory (DM), LongTermMemory (LTM), and TermMemory (TM). The DM serves as a foundational reference repository that establishes connections with both LTM and TM, thereby facilitating enhanced efficiency in document-specific retrieval operations.

Document Registration

During the document ingestion phase, the ARAI system registers each document through its source identifier and generates a corresponding document_id within the DM framework.

Data Preprocessing

The OIF component of ARAI is designed to transform academic papers from PDF format into structured LongTermMemory (LTM) representations. The preprocessing pipeline

initiates with the ingestion of academic publications. While academic papers typically adhere to standardized structural conventions—including abstract, introduction, literature review, methodology, findings, conclusion, discussion, and limitations or future work sections—nomenclature variations exist due to publisher-specific formatting requirements. Furthermore, page-based parsing methodologies often result in sectional content fragmentation, leading to contextual incompleteness and potential misinterpretation.

To mitigate content fragmentation issues, the OIF employs the "docling" Python library to convert academic papers into comprehensive markdown text format. Subsequently, the markdown content undergoes sectional segmentation into discrete chunks. These initial chunks are then consolidated through a merging process whereby segments containing fewer than 10 words are combined with their subsequent chunks. The consolidated chunks undergo metadata extraction, capturing sectional identifiers, source names, and document types. Finally, the consolidated chunks with associated metadata are further segmented using a split-overlapping methodology with parameters of 500 maximum tokens and 150-token overlap, yielding optimally-sized chunks prepared for subsequent processing stages.

Knowledge Storage Processes

Following the chunking of academic papers, ARAI initiates two concurrent processes: chunk enrichment with subsequent embedding and storage in LongTermMemory (LTM), and term extraction with storage in TermMemory (TM).

The enrichment process involves processing each chunk through a ContextEnricher Agent with AWS Bedrock meta.llama3-2-11b-instruct-v1:0, which generates a comprehensive summary of the chunk content. The resulting summary is integrated with the original context and stored in LTM across three distinct formats: raw (original content), enrich (summary), and combo (concatenated enriched and raw content). Subsequently, the

raw, enrich, and combo representations undergo embedding transformation into 1024-dimensional dense vectors utilizing the BAAI/bge-m3 embedding model from Hugging Face. Each vector is then stored in LTM with corresponding fields including raw_embedding, enrich_embedding, and combo_embedding, along with the associated document_id.

The term extraction process commences with the deployment of a TermExtractor Agent with AWS Bedrock meta.llama4-maverick-17b-instruct-v1:0, which identifies and extracts specialized terminology including technical terms, acronyms, abbreviations, frameworks, algorithms, jargon, and proper names. The agent additionally categorizes term types, identifies supporting evidence within chunks, and provides contextual explanations based on chunk content. These extracted terms are subsequently stored in DocumentMemory (DM) with their corresponding document_id.

Upon completion of both the enrichment and term extraction processes, the knowledge storage procedures conclude.

Online-Retrieval Flow (ORF)

The ORF encompasses four primary processes: query input processing, term retrieval processing, context retrieval processing, and response generation processing.

Query Input Process

This process initiates by routing user queries through a Router Agent, which orchestrates the subsequent workflow processes. The Router Agent determines whether to direct queries to the TermManager Agent, ContextRetriever, or Chat Agent. Within the scope of this research, the Router Agent functions as an experimental configuration mechanism for the study.

Term Retrieval Process

Following query routing by the Router Agent, this process commences with the identification of potential terms within the query. The TermManager Agent, utilizing AWS Bedrock meta.llama3-2-11b-instruct-v1:0, detects terms and assigns corresponding confidence scores. Each identified term undergoes full-text search within TermMemory, returning the top 10 highest-matching terms. The matching results are subsequently filtered to exclude invalid records based on specific criteria: the matching term must be non-empty, and the matching term's evidence must exceed twice the length of the potential term. All validated terms are then forwarded to the Response Generation Process with their corresponding names and detailed information.

Context Retrieval Process

Upon query routing by the Router Agent, this process begins by converting the query into a 1024-dimensional vector using the BAAI/bge-m3 embedding model. Vector search is then performed on LTM across raw_embedding, enrich_embedding, or combo_embedding fields based on configuration parameters. This operation returns the top 5 retrieved contexts, which are subsequently reranked using the cross-encoder/ms-marco-MiniLM-L6-v2 model to prioritize the most relevant contexts before forwarding to the Response Generation Process. However, all experimental evaluations in this study utilize exclusively raw_embedding for performance assessment, as the research evaluates TermManager performance against established baselines.

Response Generation Process

This final process integrates user queries, detailed terms, and contexts to construct a comprehensive prompt for generating the final user response.

For prompt construction, detailed terms can be incorporated through three methodological approaches: ‘evidence’, ‘explanation’, or ‘both’. The ‘evidence’ method constructs prompts using exclusively the evidence of terms found in TM. The ‘explanation’ method utilizes corresponding term explanations for prompt construction. The ‘both’ method incorporates both evidence and explanations into the prompt. Subsequently, contexts are integrated into the prompt before appending the user query to complete the final prompt construction.

Upon prompt completion, the Chat Agent, utilizing either AWS Bedrock `meta.llama3-2-11b-instruct-v1:0`, `meta.llama4-scout-17b-instruct-v1:0`, or `llama4-maverick-17b-instruct-v1:0` (based on configuration), generates a response and returns the result, concluding the workflow.

Dataset Construction

To evaluate the performance of ARAI (Assistant Researcher AI), a specialized dataset was manually constructed from academic papers within the artificial intelligence and machine learning domains. The dataset construction process encompassed source material selection, question and answer generation procedures, and dataset limitations.

Source Material Selection

Eighteen academic papers published between 2024 and 2025 were systematically curated to ensure that the Large Language Models employed in the experiments had not been exposed to contemporary terminology and concepts during their training phases. All selected publications were exclusively focused on AI/ML domains with which the researcher possessed domain expertise.

Question and Answer Generation Process

Thirty-three questions were manually paired with answers extracted directly from the selected academic papers. The questions were categorized into two distinct types. The first category comprises 11 acronym-related questions following the format "What does {term} stand for?" This category focuses on definitional and expansive aspects of terminology, requiring concise responses. The second category encompasses 22 explanation-related questions structured as "What is {term}?" This category emphasizes conceptual explanations and technical details, necessitating comprehensive responses.

Dataset Limitations

The dataset exhibits several limitations including size constraints, domain specificity, and language restrictions. Size constraints result from the manual curation methodology employed in dataset construction. Domain specificity is limited exclusively to AI/ML fields, resulting in reduced generalizability to other academic disciplines. Language limitations restrict the dataset to English-only content, precluding comprehensive multilingual evaluation capabilities.

This dataset serves as both training and evaluation data for assessing ARAI's performance in academic question-answering tasks, particularly in processing domain-specific terminology and generating complex explanations.

Evaluation Framework

Evaluation Metrics

The Assistant Researcher AI (ARAI) system is evaluated using ROUGE (Recall-Oriented Understudy for Gisting Evaluation) scores to quantify the overlap between

ground-truth references and system-generated predictions. While ROUGE-1 (unigram overlap) and ROUGE-2 (bigram overlap) are utilized for basic content and phrase-level assessment, these traditional n-gram metrics were de-prioritized in favor of ROUGE-L to better capture the semantic continuity required for academic explanations.

This prioritization addresses the observation that academic papers frequently suffer from "sectional content fragmentation" due to traditional page-based parsing methodologies. While the Offline-Indexing Flow (OIF) utilizes markdown conversion via the "docling" library as a primary mitigation strategy to unify document structure, ROUGE-L provides a secondary layer of robustness through its Longest Common Subsequence (LCS) algorithm. By identifying the longest co-occurring in-sequence n-grams, LCS is significantly more resilient to residual fragmentation caused by parsing boundaries compared to simple unigram or bigram overlap.

Consequently, ROUGE-L (reported as Precision, Recall, and F1) serves as the primary performance indicator for assessing the structural completeness and coherence of responses. This is particularly critical for the multi-sentence responses required in the acronym-related (definitional) and explanation-related (conceptual) task categories.

Resource Efficiency Metrics

The ARAI system evaluates computational overhead through granular token analysis to determine the Resource Return on Investment (ROI). This analysis ensures a balanced characterization of system performance by weighing accuracy gains against operational costs.

Standardized Measurement Unit

While termed "token consumption" for industry alignment, the study utilizes word counts as the fundamental unit of measurement. This approach ensures cross-model equivalence, as the different architectures—Llama 3.2 (L), Scout (S), and Maverick

(M)—employ varying tokenization algorithms that would otherwise bias the resource analysis.

Word Count Measurement

Computational cost is recorded as mean words ($\pm$ std). This metric accounts for the inherent variance in model output length and the differing densities of retrieved context across the 33-question dataset.

Variance Significance

The standard deviation (std) is utilized to distinguish the stability of standalone TermManager settings from retrieval-augmented settings. Experimental data reveals that standard deviation is significantly higher in TF and TT configurations (ranging from ± 486 to ± 492 words) due to the variable length of full-document context chunks. Conversely, standalone TermManager (FF) settings exhibit significantly lower variance (± 1 to ± 89 words).

Mathematical Definition: Token Reduction Rate (TRR)

The relative change in token consumption compared to the industry-standard baseline is expressed as the Token Reduction Rate (TRR). In all performance tables, this value is represented as the Δ change (%) column.

The TRR is calculated using the following formula:

$$TRR = \frac{W_{config} - W_{anchor}}{W_{anchor}} \times 100$$

Where:

- W_{config} is the mean word count of the specific evaluated configuration
- W_{anchor} is the mean word count of the **TF-base** configuration (1,457 words), which serves as the primary comparative anchor for the study.

Under this formula, a negative value (-) indicates resource efficiency gains (reduction in consumption), while a positive value (+) indicates the operational overhead (increase in consumption) incurred by adding terminological refinement to a retrieval flow.

Evaluation methodology

A comprehensive ablation study was conducted to evaluate individual component contributions using a factorial experimental design. The study incorporated four primary variables to systematically assess system performance across different configurations. The Models variable encompassed three variants including AWS Bedrock meta.llama3-2-11b-instruct-v1:0 (Llama), meta.llama4-scout-17b-instruct-v1:0 (Scout) and llama4-maverick-17b-instruct-v1:0 (Maverick) architectures. Term Processing was evaluated across four distinct modes comprising base (skip), evidence, explanation, and hybrid methodologies. Context Retrieval was assessed in two operational states of True and False configurations. Reranking functionality was examined in two states of True and False, with activation occurring exclusively when Context Retrieval was enabled.

Variables	Configurations	Nomenclature
Models (M)	Llama, Scout, Maverick	(L), (S), (M)
Term Processing (T)	skip, evidence, explanation, hybrid	base, evid, expl, hyb
Context Retrieval (C)	True, False	T, F

Reranking (R)	True (only when C=True), False	T, F
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Model Nomenclature

The following shorthand notation is utilized in performance tables to identify specific model architectures for comparative brevity. While certain technical analyses may refer to the smallest model variant as Llama3.2-11B, all experimental configurations utilize the meta.llama3-2-11b-instruct-v1:0 versioning.

- (L): meta.llama3-2-11b-instruct-v1:0 – Architecture serving as the high-efficiency, low-parameter baseline.
- (S): meta.llama4-scout-17b-instruct-v1:0 – Intermediate-scale architecture serving as the balanced performer for retrieval-augmented tasks.
- (M): llama4-maverick-17b-instruct-v1:0 – High-capacity architecture specialized in recall optimization and conceptual depth.

Operational Phenomena Under Evaluation

The evaluation framework is designed to detect and quantify two critical phenomena observed during terminological augmentation: the Efficiency Apex and Task Saturation.

- **Efficiency Apex:** This is defined as the operational state where a configuration achieves its maximum raw precision metrics using the minimum required token count. This phenomenon is primarily evaluated within FF (no retrieval) configurations to determine the Resource ROI of standalone terminological context.
- **Task Saturation:** This is defined as the point of diminishing returns in high-token environments where the addition of further context (retrieved chunks and terminological components) yields stagnant or negative performance growth. This is

evaluated primarily in TT (reranking) configurations to identify the capacity limits of the participating model architectures.

Baseline Comparison

ARAI performance was systematically evaluated against nine baseline configurations to establish comprehensive benchmarks across different model architectures. The baseline evaluation framework encompassed three distinct model variants, each assessed under three fundamental configuration states to ensure robust comparative analysis.

Primary Baseline Anchor

To provide a rigorous delta analysis of terminological enhancement, this study designates the "TF-base" configuration as the primary baseline anchor. This configuration represents a standard Retrieval-Augmented Generation (RAG) baseline where Context Retrieval is enabled (T), Reranking is disabled (F), and Term Processing is set to 'base'. By establishing this specific configuration as the 0.00% delta reference point for both performance metrics and token consumption, the study isolates the specific performance uplift and resource costs directly attributable to the TermManager methodologies—evidence, explanation, or both.

Item	Model	Term Processing	Context Retrieval	Rerank Processing	Configuration Designation
1	Llama (L)	skip (base)	False (F)	False (F)	FF-base
2	Llama (L)	skip (base)	True (T)	False (F)	TF-base (Primary Anchor)
3	Llama (L)	skip (base)	True (T)	True (T)	TT-base
4	Scout (S)	skip (base)	False (F)	False (F)	FF-base
5	Scout (S)	skip (base)	True (T)	False (F)	TF-base (Primary Anchor)

6	Scout (S)	skip (base)	True (T)	True (T)	TT-base
7	Maverick (M)	skip (base)	False (F)	False (F)	FF-base
8	Maverick (M)	skip (base)	True (T)	False (F)	TF-base (Primary Anchor)
9	Maverick (M)	skip (base)	True (T)	True (T)	TT-base

Base Comparison Table

The Llama model baseline configurations included an unenhanced variant with Model set to Llama, Term Processing disabled through skip mode, Context Retrieval deactivated, and Reranking functionality disabled. The second Llama baseline incorporated Context Retrieval capabilities while maintaining Term Processing in skip mode and Reranking deactivated. The third Llama baseline combined Context Retrieval with Reranking functionality while preserving the skip mode for Term Processing.

The Scout model baseline configurations followed an identical pattern, comprising an unenhanced variant with Model set to Scout, Term Processing in skip mode, Context Retrieval disabled, and Reranking deactivated. The second Scout baseline enabled Context Retrieval while maintaining Term Processing in skip mode and Reranking disabled. The third Scout baseline integrated both Context Retrieval and Reranking capabilities while retaining Term Processing in skip mode.

The Maverick model baseline configurations maintained consistency with the established pattern, featuring an unenhanced variant with Model set to Maverick, Term Processing in skip mode, Context Retrieval disabled, and Reranking functionality deactivated. The second Maverick baseline activated Context Retrieval while preserving Term Processing in skip mode and Reranking disabled. The third Maverick baseline combined Context Retrieval and Reranking functionalities while maintaining Term Processing in skip mode.

The experimental design generated thirty-six unique configuration combinations, with each combination evaluated against thirty-three questions from the manually-curated dataset. In accordance with the prioritization of sequence-based continuity and robustness against content fragmentation, performance assessment primarily reported ROUGE-L metrics (F1, Precision, and Recall) to evaluate each TermManager methodology, as it provides a more robust measure of structural completeness for academic explanations. The evaluation framework encompassed three primary comparative analyses: assessment across all thirty-three questions, focused evaluation of eleven acronym-related questions, and targeted analysis of twenty-two explanation-related questions to ensure comprehensive performance characterization across different question types.

Hardware and Computational Configurations

All experimental procedures were executed utilizing a hybrid cloud and on-premises infrastructure to optimize both performance and cost-effectiveness. The cloud-based computational resources employed AWS Bedrock models hosted in the Oregon region (us-west-2), encompassing meta.llama3-2-11b-instruct-v1:0, meta.llama4-scout-17b-instruct-v1:0, and llama4-maverick-17b-instruct-v1:0 architectures. These specific model variants were selected based on their optimal balance of processing speed and computational cost efficiency.

The on-premises infrastructure utilized PostgreSQL database systems for storing DocumentMemory (DM), LongTermMemory (LTM), and TermMemory (TM) components. PostgreSQL was selected due to its comprehensive functionality suite, including native vector search capabilities, full-text search functionality, and JSONB schema support for advanced search and filtering operations. Additionally, the BAAI/bge-m3 embedding model and cross-encoder/ms-marco-MiniLM-L6-v2 reranking model were deployed on local GPU

hardware utilizing an NVIDIA GTX 4080 graphics processing unit. This hybrid approach was implemented to minimize cloud service expenditures while maintaining computational performance standards for embedding generation and context reranking operations.

Finding

Find examples X, y_true, y_pred

All Question Analysis

ROUGE-L: acronym and explanation												
Method	F1				Precision				Recall			
	hyb	evid	expl	base	hyb	evid	expl	base	hyb	evid	expl	base
Llama3.2-11B												
FF	0.6434	0.6587	0.6054	0.1502	0.7424	0.7525	0.7249	0.1853	0.6084	0.6258	0.5666	0.1446
TF	0.655	0.6909	0.6714	0.6124	0.6928	0.6985	0.7012	0.6255	0.6626	0.7252	0.6957	0.6494
TT	0.6591	0.6959	0.6421	0.6051	0.7003	0.7046	0.6759	0.6187	0.6619	0.7226	0.6485	0.635
Maverick-17B												
FF	0.6416	0.6604	0.5969	0.1058	0.5943	0.6269	0.5663	0.1056	0.7489	0.745	0.7029	0.1245
TF	0.6509	0.6538	0.6802	0.6413	0.5712	0.5791	0.609	0.5678	0.8388	0.8332	0.8478	0.8046
TT	0.636	0.6255	0.6483	0.6515	0.5597	0.5567	0.5805	0.5789	0.8106	0.8136	0.8286	0.8121
Scout-17B												
FF	0.6143	0.6179	0.6248	0.1228	0.6119	0.5982	0.6053	0.1436	0.7058	0.7198	0.7596	0.1307
TF	0.7199	0.6945	0.705	0.665	0.7046	0.6619	0.6571	0.6008	0.7859	0.8008	0.8422	0.8094
TT	0.693	0.7059	0.6761	0.6586	0.686	0.6794	0.6344	0.613	0.755	0.7973	0.7983	0.7764

Table 1: Experimental configuration notation where FF denotes no context retrieval and no reranking functionality, TF indicates context retrieval without reranking, and TT represents context retrieval with reranking enabled. The abbreviation "base" refers to baseline configurations, "expl" denotes explanation methodology, "evid" represents evidence-based approach, and "hyb" indicates the utilization of both evidence and explanation components in term processing.

ROUGE-L: acronym and explanation												
Method	F1				Precision				Recall			
	hyb	evid	expl	base	hyb	evid	expl	base	hyb	evid	expl	base
Llama3.2-11B												
FF	328.36%	338.55%	303.06%		300.65%	306.10%	291.20%		320.75%	332.78%	291.84%	
TF	6.96%	12.82%	9.63%		10.76%	11.67%	12.10%		2.03%	11.67%	7.13%	
TT	8.92%	15.01%	6.11%		13.19%	13.88%	9.25%		4.24%	13.80%	2.13%	
Maverick-17B												
FF	506.43%	524.20%	464.18%		462.78%	493.66%	436.27%		501.53%	498.39%	464.58%	
TF	1.50%	1.95%	6.07%		0.60%	1.99%	7.26%		4.25%	3.55%	5.37%	
TT	-2.38%	-3.99%	-0.49%		-3.32%	-3.83%	0.28%		-0.18%	0.18%	2.03%	
Scout-17B												
FF	400.24%	403.18%	408.79%		326.11%	316.57%	321.52%		440.02%	450.73%	481.18%	
TF	8.26%	4.44%	6.02%		17.28%	10.17%	9.37%		-2.90%	-1.06%	4.05%	
TT	5.22%	7.18%	2.66%		11.91%	10.83%	3.49%		-2.76%	2.69%	2.82%	

Table 2: Comparative performance analysis of TermManager methodologies against baseline configurations across different model architectures and retrieval mechanisms.

Analysis of All Questions(Table 1 & 2)

The analysis of comprehensive question performance across all categories reveals a significant performance uplift when integrating TermManager methodologies compared to baseline configurations. These findings demonstrate that while the system achieves its most dramatic percentage growth in FF (no context retrieval) configurations due to lower baselines, the implementation of TF (retrieval without reranking) and TT (retrieval with reranking) configurations showcases the system's ability to maintain high absolute scores across F1, Precision, and Recall. The data suggests that the integration of terminological

context is universally beneficial, though the optimal methodology (evidence, explanation, or hybrid) varies depending on the specific model architecture and metric being targeted.

Comprehensive Performance Analysis

The experimental results indicate that the intermediate-scale model, Scout-17B, emerged as the most balanced performer in retrieval-augmented settings, achieving the highest overall F1 score. The smallest model, Llama3.2-11B, demonstrated a clear preference for evidence-based approaches, showing substantial growth and high raw precision. In contrast, the largest model, Maverick-17B, exhibited a specialized capability in recall optimization using explanation-based methodologies, though it showed signs of saturation in more complex TT configurations where growth compared to its strong baseline was minimal or slightly negative.

Metric-Specific Findings

F1 Score

The optimization was most effectively achieved by the Scout-17B model, which attained a peak score of 0.7199 when utilizing the ‘hybrid’ methodology in a TF configuration. The Llama3.2-11B model reached its performance apex of 0.6959 using the ‘evidence’ methodology in a TT configuration, while Maverick-17B achieved a peak F1 of 0.6802 using the ‘explanation’ method in a TF setting. Growth patterns for F1 were most pronounced in the FF configuration for Maverick-17B, which saw a massive 524.20% increase using the ‘evidence’ methodology, while Llama3.2-11B recorded a 338.55% increase in the same setting.

Precision

The performance reached its highest raw levels with the Llama3.2-11B model, which achieved 0.7525 using the ‘evidence’ methodology in an FF configuration. When examining retrieval-augmented growth, Scout-17B demonstrated the most significant improvement with a 17.28% increase using the ‘hybrid’ methodology in a TF configuration, reaching a raw precision of 0.7046. Llama3.2-11B maintained strong precision in TT settings with a 0.7046 score using ‘evidence,’ while Maverick-17B peaked at 0.6269 precision in the FF ‘evidence’ configuration.

Recall

The optimization showed that the larger architectures are particularly adept at capturing relevant information through descriptive context. Maverick-17B established the strongest recall performance observed in the study, attaining a peak of 0.8478 through the ‘explanation’ methodology in a TF configuration. This was closely followed by Scout-17B, which achieved its maximum recall of 0.8422 using the same ‘explanation’ and TF combination. Growth in this metric was most dramatic for Maverick-17B in the FF ‘hybrid’ configuration, which recorded a 501.53% improvement over its baseline.

Model-Specific Findings

Llama3.2-11B (Smallest)

Llama3.2-11B demonstrates a consistent preference for the ‘evidence’ methodology to maximize its performance across all evaluated metrics. While it achieves its most dramatic growth through the ‘evidence’ methodology in no-retrieval scenarios (FF), such as a 338.55% increase in F1, its peak refined performance relies on the same methodology to reach a 0.6959 F1 and 0.7226 Recall in TT settings. This indicates that smaller architectures benefit

significantly from direct evidence when processing terminological information, allowing them to remain highly competitive with larger models.

Maverick-17B (Largest)

Maverick-17B maintains high raw performance but shows signs of task saturation in its baseline, particularly in TT configurations where growth for F1 was slightly negative at -0.49%. The model performs optimally using the ‘explanation’ methodology for F1 and Recall in retrieval-augmented settings, achieving its study-wide peak recall of 0.8478 in the TF configuration. Its ability to leverage complex explanations suggests that its larger scale provides a robust foundation for processing detailed terminological context, even if its relative growth against baselines is lower than smaller models.

Scout-17B (Intermediate)

Scout-17B represents the most balanced and effective architecture for these tasks, achieving the highest overall F1 score using the ‘hybrid’ methodology in TF configurations. It consistently demonstrates flexibility by optimizing different metrics with different methodologies: using ‘hybrid’ for peak precision (0.7046) and ‘explanation’ for peak recall (0.8422) in TF settings. The model demonstrates that the intermediate scale is highly responsive to the integration of diverse terminological context, particularly when utilizing retrieval-without-reranking mechanisms.

Summary

Overall, the results demonstrate that integrating terminological context via TermManager consistently improves performance across models and metrics, with the magnitude and nature of gains depending on model scale and configuration. While FF settings show the largest relative growth due to low baselines, retrieval-augmented

configurations (TF and TT) deliver stronger and more stable absolute performance. The intermediate Scout-17B model emerges as the most balanced, achieving the highest overall F1 and adapting effectively to different methodologies, whereas the smaller Llama3.2-11B benefits most from evidence-based inputs and the larger Maverick-17B excels in recall through explanation-driven context but exhibits baseline saturation. Collectively, these findings indicate that terminological augmentation is universally beneficial, with optimal methodology selection being model- and metric-specific.

Type Question Analysis

Acronym-Related Questions

ROUGE-L: acronym												
Method	F1				Precision				Recall			
	hyb	evid	expl	base	hyb	evid	expl	base	hyb	evid	expl	base
Llama3.2-11B												
FF	0.9091	0.9495	0.8571	0.0487	0.9091	0.9432	0.8561	0.053	0.9091	0.9587	0.8636	0.0455
TF	0.9177	0.9062	0.9567	0.8364	0.9015	0.875	0.9318	0.8322	0.9545	0.9587	1	0.8442
TT	0.9177	0.886	0.8456	0.8234	0.9015	0.8659	0.8318	0.8094	0.9545	0.9132	0.8636	0.8442
Maverick-17B												
FF	0.9394	0.9382	0.8961	0.0982	0.9273	0.9264	0.8864	0.0949	1	1	0.9091	0.1302
TF	0.9622	0.9752	0.9622	0.9362	0.9383	0.961	0.9383	0.9307	1	1	1	0.9545
TT	0.9622	0.9752	0.9622	0.9752	0.9383	0.961	0.9383	0.961	1	1	1	1
Scout-17B												
FF	0.8884	0.8837	0.9055	0.0564	0.8663	0.8668	0.8913	0.0575	1	0.9587	1	0.0819
TF	1	0.9495	0.987	0.9495	1	0.9432	0.9773	0.9432	1	0.9587	1	0.9587
TT	0.987	0.9495	0.987	0.9004	0.9773	0.9432	0.9773	0.8947	1	0.9587	1	0.9132

Table 3: Absolute Performance for Acronym-Related Questions (ROUGE-L)

ROUGE-L: acronym												
Method	F1				Precision				Recall			
	hyb	evid	expl	base	hyb	evid	expl	base	hyb	evid	expl	base
Llama3.2-11B												

FF	1766.74%	1849.69%	1659.96%	1615.28%	1679.62%	1515.28%	1898.02%	2007.03%	1798.02%
TF	9.72%	8.35%	14.38%	8.33%	5.14%	11.97%	13.07%	13.56%	18.46%
TT	11.45%	7.60%	2.70%	11.38%	6.98%	2.77%	13.07%	8.17%	2.30%
Maverick-17B									
FF	856.62%	855.40%	812.53%	877.13%	876.19%	834.04%	668.05%	668.05%	598.23%
TF	2.78%	4.17%	2.78%	0.82%	3.26%	0.82%	4.77%	4.77%	4.77%
TT	-1.33%	0.00%	-1.33%	-2.36%	0.00%	-2.36%	0.00%	0.00%	0.00%
Scout-17B									
FF	1475.18%	1466.84%	1505.50%	1406.61%	1407.48%	1450.09%	1121.00%	1070.57%	1121.00%
TF	5.32%	0.00%	3.95%	6.02%	0.00%	3.62%	4.31%	0.00%	4.31%
TT	9.62%	5.45%	9.62%	9.23%	5.42%	9.23%	9.51%	4.98%	9.51%

Table 4

Analysis of Acronym-Related Questions (Table 3 & 4)

The analysis of acronym-related questions reveals near-optimal performance across all evaluated model architectures, particularly when integrated with TermManager methodologies. These findings demonstrate that acronym definition and expansion tasks represent a domain where the system achieves its highest absolute scores, frequently reaching perfect benchmarks in precision and recall. While the FF (no context retrieval) configuration yields the most dramatic percentage growth due to exceptionally low baseline performance, the TF (retrieval without reranking) and TT (retrieval with reranking) configurations provide a clearer view of how specific methodologies refine system accuracy.

Comprehensive Performance Analysis

The experimental results indicate that the intermediate-scale model, Scout-17B, emerged as the superior performer for acronym-related tasks, achieving perfect 1.0 scores across several metrics. The smallest model, Llama3.2-11B, demonstrated the most substantial relative growth against baselines, particularly within the 'explanation' methodology. In contrast, the largest model, Maverick-17B, while maintaining high raw scores, exhibited

stagnant or negative growth in more complex retrieval settings like TT, suggesting that its baseline already possesses strong inherent knowledge for this specific task.

Metric-Specific Findings

F1 Score

The optimization was most effectively achieved by the Scout-17B model, which attained a perfect score of 1.0 when utilizing the 'hybrid' methodology in a TF configuration. The Llama3.2-11B model reached its performance apex of 0.9567 using the 'explanation' methodology in a TF configuration, while Maverick-17B achieved a peak F1 of 0.9752 using the 'evidence' method in both TF and TT settings. Growth patterns for F1 were most pronounced in the FF configuration for Llama3.2-11B, which saw a massive 1849.69% increase using the 'evidence' methodology.

Precision

The performance followed a similar trajectory, with Scout-17B achieving a perfect 1.0 precision score in the TF configuration with 'hybrid' methodology. Llama3.2-11B exhibited high accuracy with a raw precision of 0.9432 in the FF configuration using the 'evidence' methodology, representing a 1679.62% improvement over its baseline. Maverick-17B peaked at 0.961 precision using the 'evidence' methodology in TT and TF configurations, though its growth in the TT setting remained stagnant at 0.00% compared to its baseline.

Recall

The optimization showed the highest frequency of perfect results across the study. Llama3.2-11B attained a perfect 1.0 recall score using the 'explanation' methodology in the TF configuration. Both Scout-17B and Maverick-17B consistently achieved perfect recall scores across multiple configurations; Scout-17B maintained 1.0 recall across FF, TF, and TT

configurations using both 'hybrid' and 'explanation' methodologies. Llama3.2-11B recorded the highest growth in this metric with a 2007.03% recall increase in the FF 'evidence' configuration.

Model-Specific Findings

Llama3.2-11B (Smallest)

Llama3.2-11B demonstrates a clear preference for the 'explanation' methodology to maximize its performance when retrieval is enabled, particularly for F1 and Recall. While it achieves its most dramatic growth through the 'evidence' methodology in no-retrieval scenarios (FF), its peak refined performance relies on 'explanation' to reach a 0.9567 F1 and 1.0 Recall in TF settings. This indicates that smaller architectures benefit significantly from descriptive context when handling acronym-related processing.

Maverick-17B (Largest)

Maverick-17B maintains high raw performance but shows signs of task saturation in its baseline, particularly in TT configurations where growth often dropped to 0.00% or became negative (-2.36% in precision for 'hybrid' and 'explanation' methodologies). The model performs optimally using the 'evidence' methodology for F1 and precision in retrieval-augmented settings. Its recall remains consistently perfect across most TermManager configurations, suggesting the model scale provides robust foundational knowledge for acronym tasks regardless of the specific methodology applied.

Scout-17B (Intermediate)

Scout-17B represents the most balanced and effective architecture for acronym tasks, achieving perfect scores in F1 and Precision through the 'hybrid' methodology in TF configurations. It consistently outperforms larger and smaller models in raw score metrics

when utilizing retrieval. The model demonstrates flexibility by maintaining perfect recall across all retrieval settings using both 'hybrid' and 'explanation' methodologies, indicating that the integration of diverse terminological context is highly effective for this scale of architecture.

Summary

Acronym-related tasks represent the strongest performance domain across all models, with TermManager integration frequently yielding near-perfect or perfect precision, recall, and F1 scores. While FF configurations show extreme relative gains due to minimal baselines, TF and TT settings better illustrate refined accuracy, particularly for the intermediate Scout-17B model, which consistently achieves perfect scores and emerges as the most reliable architecture for this task. The smaller Llama3.2-11B benefits most from explanation-based context when retrieval is enabled, whereas the larger Maverick-17B shows clear signs of baseline saturation, maintaining high raw performance but limited growth. Overall, these results indicate that acronym processing is a mature capability for modern models, with retrieval-enhanced terminological context primarily serving to stabilize and perfect already strong performance.

Explanation-Related Questions

ROUGE-L: explanation												
Method	F1				Precision				Recall			
	hyb	evid	expl	base	hyb	evid	expl	base	hyb	evid	expl	base
Llama3.2-11B												
FF	0.5106	0.5133	0.4796	0.201	0.659	0.6571	0.6593	0.2515	0.4581	0.4594	0.418	0.1941
TF	0.5236	0.5833	0.5287	0.5004	0.5885	0.6102	0.5859	0.5222	0.5167	0.6084	0.5435	0.552

TT	0.5298	0.6008	0.5404	0.4959	0.5997	0.6239	0.598	0.5234	0.5156	0.6273	0.5409	0.5305
Maverick-17B												
FF	0.4927	0.5216	0.4473	0.1096	0.4278	0.4772	0.4062	0.111	0.6233	0.6176	0.5998	0.1216
TF	0.4952	0.4931	0.5393	0.4938	0.3877	0.3882	0.4443	0.3863	0.7582	0.7498	0.7716	0.7296
TT	0.4729	0.4506	0.4913	0.4897	0.3704	0.3545	0.4016	0.3878	0.7159	0.7204	0.7428	0.7181
Scout-17B												
FF	0.4773	0.4851	0.4845	0.1559	0.4847	0.4638	0.4622	0.1867	0.5587	0.6003	0.6394	0.1551
TF	0.5798	0.5671	0.564	0.5227	0.5569	0.5213	0.497	0.4296	0.6789	0.7218	0.7633	0.7348
TT	0.5459	0.5842	0.5206	0.5376	0.5404	0.5475	0.463	0.4721	0.6325	0.7165	0.6975	0.708

Table 5: Absolute Performance for Explanation-Related Questions (ROUGE-L).

ROUGE-L: explanation												
Method	F1				Precision				Recall			
	hyb	evid	expl	base	hyb	evid	expl	base	hyb	evid	expl	base
Llama3.2-11B												
FF	154.03%	155.37%	138.61%		162.03%	161.27%	162.15%		136.01%	136.68%	115.35%	
TF	4.64%	16.57%	5.66%		12.70%	16.85%	12.20%		-6.39%	10.22%	-1.54%	
TT	6.84%	21.15%	8.97%		14.58%	19.20%	14.25%		-2.81%	18.25%	1.96%	
Maverick-17B												
FF	349.54%	375.91%	308.12%		285.41%	329.91%	265.95%		412.58%	407.89%	393.26%	
TF	0.28%	-0.14%	9.21%		0.36%	0.49%	15.01%		3.92%	2.77%	5.76%	
TT	-3.43%	-7.98%	0.33%		-4.49%	-8.59%	3.56%		-0.31%	0.32%	3.44%	
Scout-17B												
FF	206.16%	211.16%	210.78%		159.61%	148.42%	147.56%		260.22%	287.04%	312.25%	
TF	10.92%	8.49%	7.90%		29.63%	21.35%	15.69%		-7.61%	-1.77%	3.88%	
TT	1.54%	8.67%	-3.16%		14.47%	15.97%	-1.93%		-10.66%	1.20%	-1.48%	

Table 6

Analysis of Explanation-Related Questions (Table 5 & 6)

The analysis of explanation-related questions indicates that the implementation of TermManager methodologies across varying model architectures consistently enhances performance compared to baseline configurations. These improvements are particularly evident in scenarios where context retrieval is not utilized (FF), though the integration of retrieval (TF) and reranking (TT) mechanisms provides a more sophisticated understanding of how these methodologies enhance active retrieval systems.

Comprehensive Performance Analysis

The experimental evaluation reveals that the smallest model, Llama3.2-11B, generally achieved the highest overall results in F1 and Precision metrics. In contrast, the largest model, Maverick-17B, demonstrated a specialized capability in Recall optimization, particularly when utilizing retrieval-based configurations. Scout-17B maintained a balanced and flexible performance profile, showing significant precision gains in retrieval-without-reranking scenarios.

Metric-Specific Findings

F1 Score

The F1 score analysis demonstrates that the Llama3.2-11B model attained its peak performance of 0.6008 using the ‘evidence’ methodology in a TT configuration, which also represented a 21.15% gain over its retrieval baseline. Scout-17B followed this pattern by reaching its highest F1 score of 0.5842 through the same ‘evidence’ methodology and TT combination, while Maverick-17B optimized its F1 apex at 0.5393 using the ‘explanation’ methodology in a TF configuration, yielding a 9.21% improvement. These results suggest

that while the ‘evidence’ methodology drives F1 for smaller models, the ‘explanation’ methodology is more effective for the largest architecture.

Precision

Precision performance reached its highest raw levels with the Llama3.2-11B model, which achieved 0.6593 using the ‘explanation’ methodology in an FF configuration. When examining retrieval-augmented growth, Scout-17B demonstrated the most significant improvement with a 29.63% increase using the ‘hybrid’ methodology in a TF configuration, while Llama3.2-11B reached 19.20% growth using ‘evidence’ methodology in a TT setting. Maverick-17B saw its highest retrieval-based precision gain of 15.01% using the ‘explanation’ methodology in a TF scenario.

Recall

The Maverick-17B model established the strongest recall performance observed in the study, attaining a peak of 0.7716 through the ‘explanation’ methodology in a TF configuration. This was closely followed by Scout-17B, which achieved its maximum recall of 0.7633 using the ‘explanation’ methodology and TF combination. Llama3.2-11B maximized its recall performance at 0.6273 by utilizing the ‘evidence’ methodology in a TT configuration, representing an 18.25% improvement over its retrieval-based baseline.

Model-Specific Findings

Llama3.2-11B (Smallest)

This model proved to be highly efficient, consistently favoring the ‘evidence’ methodology across F1, Precision, and Recall metrics, particularly when integrated with reranking (TT) or standard retrieval (TF). It demonstrated the most robust and consistent

improvements across all retrieval-based metric categories, specifically dominating in F1 and Precision raw scores.

Maverick-17B (Largest)

The largest model showed a clear dependency on the ‘explanation’ methodology to drive performance, particularly for F1 and Recall in TF and TT scenarios. While it exhibited the highest percentage growth in FF configurations due to exceptionally low baseline levels, its specialized strength lies in recall optimization, where it significantly outperformed the smaller architectures.

Scout-17B (Intermediate)

Scout-17B exhibited a balanced and flexible profile, optimizing its performance across different methodologies depending on the specific metric targeted. It reached its peak F1 through the ‘evidence’ methodology in TT configurations, optimized precision using the ‘hybrid’ methodology in TF settings, and maximized recall via the ‘explanation’ methodology in TF configurations.

Summary

Explanation-related questions show consistent performance gains from TermManager integration across all models, with the most meaningful insights emerging in retrieval-augmented settings. The smaller Llama3.2-11B model excels in F1 and precision, driven primarily by the ‘evidence’ methodology, while the largest Maverick-17B demonstrates a clear advantage in recall through the ‘explanation’ methodology. Scout-17B maintains a balanced and adaptable profile, optimizing different metrics with different methodologies depending on the retrieval configuration. Overall, the results highlight that explanation tasks benefit strongly from model–method alignment, where the ‘evidence’

methodology favors smaller architectures and the ‘explanation’ methodology better leverages larger model capacity.

Token (Word) Consumption and Resource Analysis

Token Consumption and Performance

Configuration	Methodology	Mean tokens ($\pm$ std)	Δ change (%)	F1	Precision	Recall
FF	hybrid	223 $\pm$ 89	-84.69%	0.6434 (L)	0.7424 (L)	0.7489 (M)
	evidence	99 $\pm$ 38	-93.21%	0.6604 (M)	0.7525 (L)	0.745 (M)
	explanation	111 $\pm$ 42	-92.38%	0.6248 (S)	0.7249 (L)	0.7596 (S)
	base	44 $\pm$ 1	-96.98%	0.1502 (L)	0.1853 (L)	0.1446 (L)
TF	hybrid	1675 $\pm$ 489	14.96%	0.7199 (S)	0.7046 (S)	0.8388 (M)
	evidence	1552 $\pm$ 486	6.52%	0.6945 (S)	0.6985 (L)	0.8332 (M)
	explanation	1566 $\pm$ 490	7.48%	0.705 (S)	0.7012 (L)	0.8478 (M)
	base (anchor)	1457 $\pm$ 492	0.00%	0.665 (S)	0.6255 (L)	0.8094 (S)
TT	hybrid	1677 $\pm$ 487	15.10%	0.693 (S)	0.7003 (L)	0.8106 (M)
	evidence	1552 $\pm$ 486	6.52%	0.7059 (S)	0.7046 (L)	0.8136 (M)
	explanation	1566 $\pm$ 490	7.48%	0.6761 (S)	0.6759 (L)	0.8286 (M)
	base	1457 $\pm$ 492	0.00%	0.6586 (S)	0.6187 (L)	0.8121 (M)

Table 7: L: Llama3.2-11B; M: Maverick-17B; S: Scout-17B. The "mean tokens ($\pm$ std)" column represents standardized word counts to ensure equivalent comparison across different model tokenizers.

Analysis of Token Consumption and Performance (Table 7)

The ARAI system demonstrates a significant trade-off between operational cost and model effectiveness. While the FF (no retrieval) configurations provide a massive reduction in token consumption (between -84.69% and -96.98%), the TF and TT (retrieval-augmented)

configurations stabilize absolute performance scores at the cost of a +6.52% to +15.10% increase in token overhead. This confirms that targeted terminological context is highly efficient, while full document retrieval is more resource-intensive.

Comprehensive Performance Analysis

The findings reveal an "efficiency apex" where specific models achieve peak metrics with minimal resources. Smaller models like Llama3.2-11B reach their precision ceiling in low-token environments (-93.21% tokens), whereas intermediate and larger models like Scout-17B and Maverick-17B effectively scale with higher token counts in retrieval-augmented settings. However, a "saturation point" is observed in TT configurations, where the maximum token cost (+15.10%) does not always yield a performance return, particularly for Maverick-17B.

Metric-Specific Findings

F1 Score

The optimization was most effectively achieved by the Scout-17B model, which attained a study-wide peak score of 0.7199 in the TF 'hybrid' configuration, requiring a +14.96% token increase. Maverick-17B demonstrated high efficiency by achieving a competitive 0.6604 F1 in the FF 'evidence' setting while reducing token consumption by -93.21%.

Precision

The performance reached its highest raw levels with the Llama3.2-11B model, which achieved 0.7525 using the 'evidence' methodology in an FF configuration. This indicates that

the most precise model in the study is also among the most token-efficient, operating at a -93.21% reduction compared to the baseline.

Recall

The optimization showed that larger architectures thrive on higher token counts. Maverick-17B established the strongest recall in the study (0.8478) using the ‘explanation’ methodology in a TF configuration, which added +7.48% to the token count.

Model-Specific Findings

Model scale determines how effectively an architecture translates increased token counts into performance gains. Smaller architectures benefit from targeted, low-token evidence, while larger architectures require dense explanatory context to maximize their recall capabilities.

Llama3.2-11B (Smallest)

This model is the most resource-efficient, consistently delivering top-tier precision scores in FF settings with token reductions of over 90%. It demonstrates that smaller models do not necessarily need high-token retrieval to remain competitive.

Maverick-17B (Largest)

While expert at recall, this model shows baseline saturation in high-token settings. Despite consuming the maximum +15.10% token overhead in TT ‘hybrid’, its performance can become stagnant or negative compared to leaner TF configurations.

Scout-17B (Intermediate)

This model represents the most balanced performer, effectively leveraging the higher token counts of the TF ‘hybrid’ (+14.96%) configuration to secure the highest overall F1 score.

Summary

Overall, the integration of TermManager provides a tunable balance between cost and accuracy. FF configurations represent the highest Resource ROI, achieving massive efficiency gains (up to -93.21% tokens) while maintaining high precision. TF configurations represent the Performance Apex, where models like Scout-17B utilize a modest +14.96% token increase to achieve peak absolute scores. These findings confirm that terminological augmentation is universally beneficial, but optimal deployment requires matching the methodological token footprint to the specific capacity of the model and the priority of the task.

Final Summary

Overall, experimental results demonstrate that integrating terminological context through TermManager improves performance across all evaluated models and metrics. The magnitude of improvement varies by model scale, task type, and retrieval configuration. While FF configurations show the largest relative gains—such as Maverick-17B’s 524.20% increase in F1 due to low baselines—TF and TT configurations produce the highest and most stable absolute scores.

Model-specific analysis indicates that optimal methodological alignment depends on scale. Scout-17B achieved the highest overall F1 score (0.7199) under the ‘hybrid’ methodology in a TF configuration. Llama3.2-11B attained the highest precision (0.7525)

under the ‘evidence’ methodology, whereas Maverick-17B reached the highest recall (0.8478) but exhibited performance plateauing in TT configurations, suggesting the onset of task saturation under increased contextual integration.

Task-level evaluation reveals that acronym-related questions consistently achieve near-ceiling performance across configurations, whereas explanation-based tasks demonstrate greater variability and stronger dependence on model–method alignment.

Resource analysis shows that FF configurations achieve the highest token efficiency, reducing consumption by up to 93.21% while maintaining competitive precision. TF configurations deliver peak absolute performance at a token overhead of approximately 14.96%. Variance analysis further indicates lower performance dispersion in standalone terminological settings compared to retrieval-based configurations.

Discussion (use RQ to explain)

The following discussion chapter provides a formal analytical synthesis of the results derived from the Assistant Researcher AI (ARAI) system, interpreting the operational phenomena of Efficiency Apex and Task Saturation. This analysis evaluates how terminological context influences performance across varying model scales and configurations.

Architecture-Specific Heuristics: Parameter-Method Optimization

The effectiveness of the ARAI system is predicated on "Parameter-Method Optimization," a heuristic where model capacity dictates the ideal TermManager methodology—whether providing ‘evidence’, ‘explanation’, or ‘hybrid’ methodologies.

Llama3.2-11B (Small Scale)

This architecture demonstrates a consistent general preference for the evidence-based methodology to maximize performance across broader metrics. Smaller architectures appear to benefit significantly from direct evidence strings which provide a stable factual anchor, allowing the 11B model to reach a peak F1 of 0.6959 in TT settings and a study-wide high precision of 0.7525 in FF settings. However, the model’s methodology preference is notably task-dependent; for acronym-related questions, it reaches its performance apex (0.9567 F1 and perfect 1.0 Recall) using the ‘explanation’ methodology in TF settings. This indicates that while evidence serves as the most reliable stabilizer for general synthesis, the model requires descriptive explanatory context to maximize its accuracy for concise definitional tasks.

Scout-17B (Intermediate Scale)

Emerging as the "balanced performer," Scout-17B successfully navigates the trade-off between factual brevity and conceptual depth. It achieved the highest overall F1 score in the study (0.7199) using the 'hybrid' methodology in TF configurations. The model's reliability as the study's most effective architecture for retrieval tasks is further evidenced by its performance in acronym-related questions, where it attained perfect 1.0 benchmarks across F1, Precision, and Recall when utilizing the 'hybrid' methodology. This indicates that intermediate-scale models possess sufficient capacity to synthesize diverse terminological context without reaching a saturation point in standard retrieval flows, demonstrating a high degree of responsiveness to the integration of complex terminological data.

Maverick-17B (Large Scale)

This architecture functions as a specialized recall engine. It leverages the 'explanation' methodologies to handle high-complexity conceptual requirements, achieving the study's peak recall of 0.8478. Its scale allows it to process detailed terminological context to maximize information capture, even when relative growth appears more modest than smaller models due to high baseline performance.

The "Efficiency Apex": Maximizing Resource ROI

The Efficiency Apex represents the configuration where the system achieves maximum precision metrics using the minimum required token count. This phenomenon is most pronounced in FF (No Context Retrieval) settings, where the TermManager functions as a standalone enhancement.

The FF-evidence configuration achieved a uniform -93.21% Token Reduction Rate (TRR) relative to the TF-base anchor (1,457 words) while maintaining the study's peak raw

precision level of 0.7525 with the Llama3.2-11B. These findings confirm that the most precise model in the study is also the most token-efficient, providing an optimal balance for resource-constrained tasks.

Beyond raw word counts, this apex is characterized by a highly predictable deployment profile. Standalone TermManager (FF) settings exhibit significantly lower variance (± 1 to ± 89 words) compared to retrieval-augmented settings (± 486 to ± 492 words), where word counts fluctuate drastically due to the variable length of full-document context chunks. Specifically, the FF-evidence configuration maintains a standard deviation of only ± 38 words, ensuring consistent latency and computational costs. For production environments prioritizing cost-efficiency, latency stability, and predictable resource allocation, the FF-evidence configuration offers the highest Resource Return on Investment (ROI) by strategically matching the methodological footprint to the model's capacity.

Task Saturation and Diminishing Returns

Task Saturation occurs when the volume of context—including retrieved document chunks, reranking logic, and terminological detail—exceeds a model's capacity for effective synthesis, leading to stagnant or negative performance growth.

This is most evident in the TT (Retrieval with Reranking) configurations for the Maverick-17B model. Despite an increase in computational overhead to +15.10% in the TT-hybrid configuration, Maverick-17B exhibited significant negative growth in F1 (-2.38%) and precision (-3.32%) within the comprehensive All Question analysis. Even specialized configurations showed signs of decline, with the TT-explanation setting resulting in a -0.49% F1 growth overall, while acronym-related tasks saw precision drop by -2.36% across both 'hybrid' and 'explanation' methodologies. Theoretically, this degradation is driven by contextual redundancy. When high-capacity models are presented with overlapping

information from both retrieved chunks and terminological context, the signal-to-noise ratio decreases, hindering the attention mechanism's ability to isolate specific answers.

Domain-Specific Task Analysis: Acronyms vs. Explanations

Acronym-Related Tasks (Stabilization)

Acronym processing (e.g., "What does {term} stand for?") represents a mature capability. The ARAI system frequently achieves perfect 1.0 scores in precision and recall for these tasks. Here, the TermManager serves as a stabilization layer, ensuring that even if internal model knowledge is dated, the provided terminological context guarantees near-perfect accuracy.

Explanation-Related Tasks (Performance Driver)

These tasks are highly sensitive to Model-Method Alignment. Unlike simple acronyms, conceptual synthesis requires alignment between the model's capacity and the complexity of the input. While smaller models like Llama3.2-11B maximize effectiveness through direct evidence (0.6008 F1), the larger Maverick-17B requires explanatory context to drive its specialized recall capabilities.

Conclusion: The ARAI Tunable Framework

The ARAI framework is a tunable system that allows for the strategic balancing of computational cost against structural accuracy. While the integration of terminological context is universally beneficial, its application must be navigated between two distinct operational poles:

- The Efficiency Apex (FF-evidence): Optimal for resource-constrained environments that prioritize maximum precision and predictable resource allocation over comprehensive retrieval.
- The Performance Apex (TF-both): Recommended for comprehensive academic accuracy where the intermediate-scale Scout-17B serves as the ideal balanced performer, utilizing diverse terminological context to maximize F1 scores.

Researchers must calibrate the framework's methodological footprint based on the specific computational budget and task complexity, acknowledging that high-capacity architectures like Maverick-17B, while superior in recall, remain susceptible to Task Saturation in high-token environments.

Conclusion (conclude our own interpretation)

add this later after finishing literature review

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